LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to main plane, aero dynamic and structural considerations, loads on wings, basic structural requirements, role of wings as stru
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to main plane, aero dynamic and structural considerations, loads on wings, basic structural requirements, role of wings as structure, positioning of components, form of wing construction
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of wing structural diagrams; Safety: No physical handling of aircraft structures; Instructor supervision: Monitor diagram interpretation accuracy
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the main plane?
(b) Name two types of loads acting on a wing.
(c) How does the wing contribute to the overall structural integrity?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Aerodynamic and structural considerations • Types of loads on wings • Basic structural requirements • Role of wings as structure • Positioning of components	L & Ds	PPT, IB	15 min
(b)	• Form of wing construction • Spars and ribs function • Skin and stringer roles • Fuel tank integration • Control surface attachment	L & Ds	PPT, PDF	20 min
(c)	• Summary of wing structural elements • Key load distribution points • Material selection criteria • Maintenance implications • Final review of objectives	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how lift generation relates to wing structural strength.
(b) Describe the role of spars in resisting bending moments.
(c) Identify three critical components in wing construction and their functions.

12. Home Assignment and Follow Up	Prepare a brief note on how the wing structure supports the landing gear attachment points, referencing the next lesson on Empennage structure.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Empennage structure, configuration of longitudinal stabilizer and concord and fore plane, layout of configuration of lateral stabilizer and control	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min